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China Battery & Materials

Copper foil supply tightness resolved post price hike; October production outlook

CATL A/H shares have fallen 18% MTD (vs. CSI 300 -6%) amid concerns over slowing demand, competition, and production cuts. In our 2-Sep production outlook [note](#), we highlighted copper foil supply tightness as a near-term constraint. Since then, investors have increasingly focused on the possibility of further production cuts. These concerns were reinforced by ZE Consulting's mid-month update on 15 September, which lowered CATL's September production estimate by 12%, and again in the latest update where September production estimates were further cut by 3%. These revisions brought CATL's September production down from +9% m/m to -5% m/m. Our checks suggest copper foil price negotiations were completed in late September, implying that the supply bottleneck has largely eased. ZE Consulting expects October production to resume sequential growth (+6% m/m, back to August levels). Based on current trends, we estimate CATL will exit FY26 with >1.1TWh of production volume and ~1.0TWh of sales volume (~50% y/y growth). **CATL remains our top pick in China's battery value chain.**

- **CATL's September production estimates were revised down by a cumulative 14% from ZE Consulting's initial forecast.** In its mid-month update released after market close on 15 September, ZE Consulting cut CATL's September production estimate by 12%, followed by a further 3% reduction in its latest update, bringing expected September output growth from +9% m/m to -5% m/m. The downward revisions were attributed to tight copper foil supply, as CATL was reportedly still negotiating pricing with copper foil suppliers, while most other battery makers had already accepted higher processing fees in August. Based on our channel checks, CATL reached an agreement in late September, with processing fees increasing by Rmb1-2k/t, equivalent to only ~0.1-0.2% of battery costs. With the supply bottleneck now largely resolved, ZE Consulting expects CATL's October production to rebound 6% m/m, returning to August levels. Furthermore, according to ZE Consulting, CATL's raw material procurement volumes are expected to accelerate from mid-October, further supporting a recovery in production momentum.
- **10M26 production by major battery makers is expected to grow ~55% y/y, accelerating from 40% y/y in 1Q26 and 60% in 2Q26 to 62% in 3Q26.** The YTD growth is driven by a combination of strong underlying trends: **(1) Solid ESS demand:** Chinese players recorded 95% y/y growth in 8M26 ESS battery shipments, according to ICCSino. **(2) Continued strength in EV exports,** which rose nearly 130% y/y in the 8M26, while European EV sales grew >40% y/y over the same period. **(3) Robust China commercial EV sales,** led by heavy-duty trucks: Commercial EV car volumes increased 45% y/y, while battery volumes grew ~70% y/y in 8M26. On the other hand, **(4) China passenger EV retail sales were softer than expected,** with an expected 12% y/y decline in 9M26, per CPCA estimates. However, the weakness in domestic EV sales was largely offset by **rising battery content per vehicle,** which increased by 9 kWh per car, or ~21% y/y, in 8M26 (Figure 8).
- **Tier 2 battery makers are catching up as new capacity ramps, although CATL continues to lead.** CATL's production grew 66% y/y in 3Q26, versus

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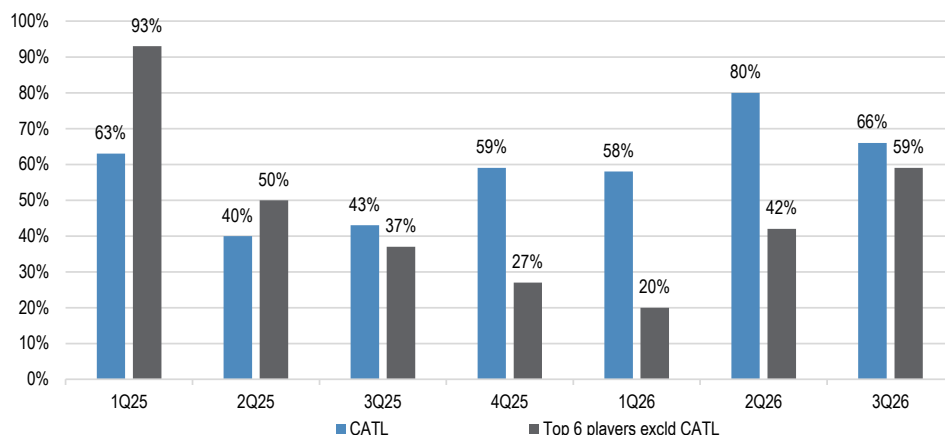
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59% for the other top-six producers excluding CATL, narrowing the gap from 38ppts in 2Q26 to 7ppts in 3Q26. The trend is set to continue in October, with EVE and CALB planning the strongest sequential growth at 11% and 19% m/m, respectively, while CATL is expected to recover by 6% m/m as the copper-foil constraint eases. We view the narrowing gap as reflecting tier 2 capacity ramp-ups and CATL's temporary September supply disruption rather than a structural loss of competitiveness, although we believe concerns over potential market-share normalization may have weighed on recent investor sentiment and contributed to CATL's recent share price weakness.

- Flattish to slightly declining October lithium production expected.** ZE Consulting expects China's lithium carbonate output to decline slightly ~1% m/m in October, as production additions and Zimbabwean ore arrivals are offset by maintenance-led cuts, leaving overall output broadly unchanged. Lithium spot and futures prices continued to soften to Rmb132k/t and Rmb119k/t, respectively, with the forward curve shifting into backwardation. Sentiment has weakened on expectations of a looser S/D balance and positioning pressure. On supply, Zimbabwean port arrivals and Zijin Manono's ramp-up (40kt production in 1H26; 120kt FY target) point to upside additions, while CATL's Jianxiawo restart timeline remains the key uncertainty. On demand, the battery consumption-tax implementation on 1 Sep weighs on expectations, with downstream restocking momentum the main swing factor to monitor. September weekly output rose to 25–26kt, vs. ~23kt levels in August. Diverging inventory trends in the week of 24 Sep also cast doubt on whether destocking has ended: newly released sample data (Sep) showed a 5kt WoW drawdown to 153kt, while the old sample (May) increased 2kt to 72kt and the original old sample restocked 5kt to 64kt—the latter two marking an inflection after ~4 months of consecutive destocking. Looking into 2027, lithium sentiment has softened as the market focuses more on the sustainability and pace of ESS demand growth. Hence, **we expect near-term lithium prices to remain under pressure.**
- Selective price increases have emerged across the battery value chain.** In 3Q26 to date, lithium carbonate and lithium hydroxide prices declined by 10% and 14%, while cathode prices fell by 8–11%. In contrast, strong downstream demand and tight supply drove copper-foil processing fees up by ~Rmb1–2k/t, or 4–10%, with a larger increase for 6μm than 5μm foil. Electrolyte prices rose by 12–18%. EV battery prices remained stable, while ESS battery prices increased 2.7% m/m in August, which we believe mainly reflected pass-through of the 2% consumption tax effective from 1 September.

Figure 1: Historical y/y growth of CATL vs. peers (y/y %)

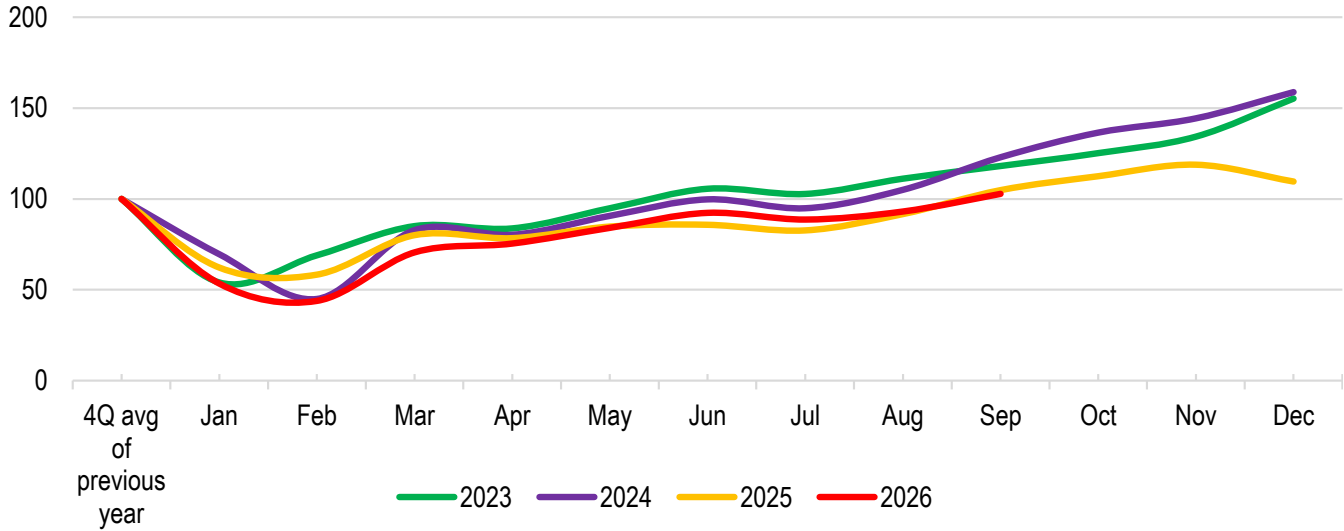


Source: ZE Consulting, J.P. Morgan. Note: Based on battery production data of top 6 battery makers: BYD, CATL, Gotion, EVE, CALB, REPT.

NEV sales and battery production trends

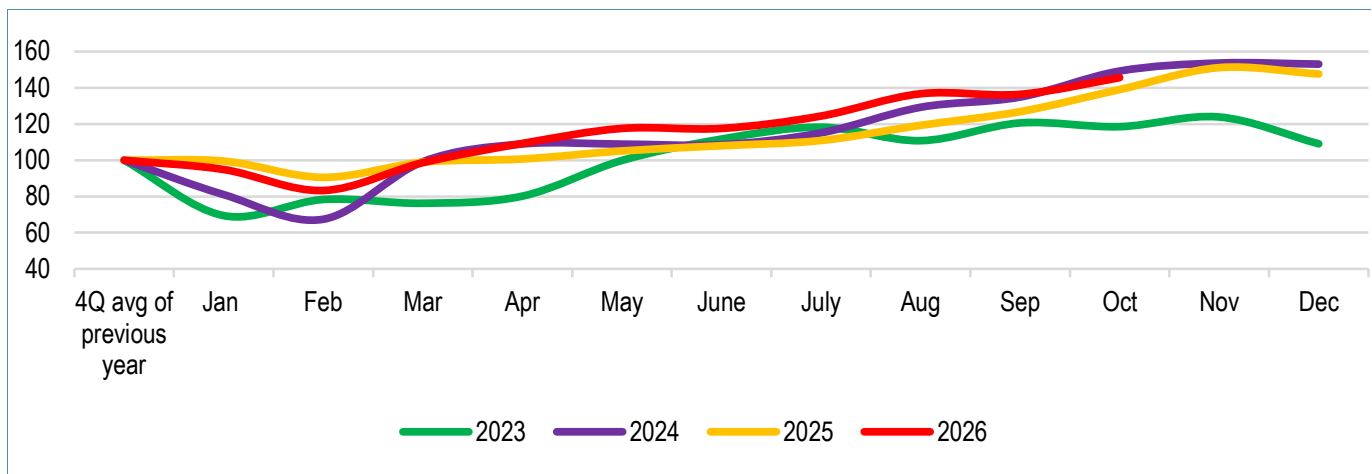
Figure 2: Historical China NEV sales monthly volume vs. previous-year 4Q average monthly sales (assumes 4Q average monthly sales volume = 100)

Est Sep-26 NEV sales up 10% m/m, trending lower than Sep-25 (up 14% m/m)



Source: CAAM, CPCA, J.P. Morgan

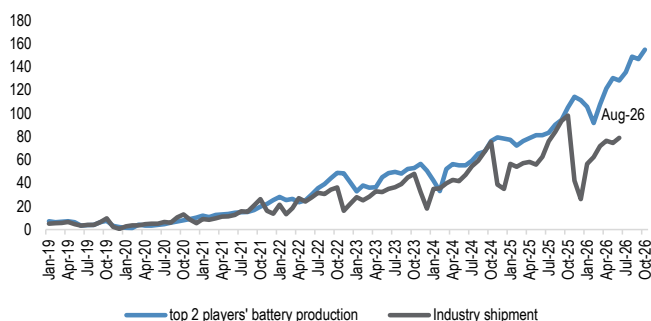
Figure 3: Historical battery production monthly volume vs. previous-year 4Q average monthly production volume (assumes 4Q average monthly production volume = 100)



Source: ZE Consulting, J.P. Morgan. Note: Based on battery production data of top 6 battery makers.

Figure 4: Top 2 battery makers' production volume vs. one-month lagged China NEV retail sales trend (GWh)

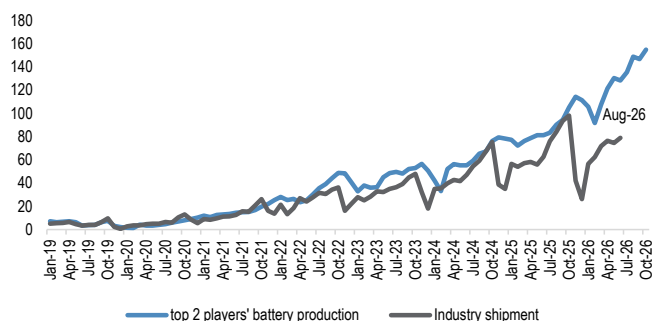
Battery production is usually one to two months ahead of NEV installation/shipments



Source: ZE Consulting, CABIA, J.P. Morgan

Figure 5: Top 2 battery makers' production volume vs. two-month lagged China NEV retail sales trend (GWh)

Battery production is usually one to two months ahead of NEV installation/shipments



Source: ZE Consulting, CABIA, J.P. Morgan

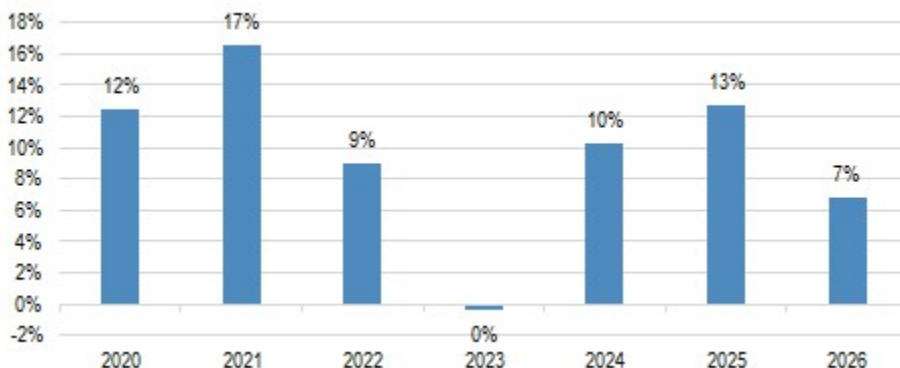
Battery production outlook

Table 1: Summary of production trends (monthly)

	Aug 25		Sep 25		Oct 25		Nov 25		Dec 25		Jan 26		Feb 26		Mar 26		Apr 26		May 26		Jun 26		Jul 26		Aug 26		Sep 26			Oct 26	
	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Y/Y	M/M	Final vs Beg Est	Y/Y	M/M
Battery maker																															
BYD	34%	8%	32%	-2%	20%	13%	10%	7%	-4%	-15%	-7%	-7%	-12%	-12%	2%	27%	3%	11%	18%	6%	27%	0%	42%	8%	44%	10%	62%	10%	5%	50%	5%
CATL	40%	9%	44%	7%	47%	11%	64%	10%	68%	2%	61%	-4%	48%	-14%	64%	14%	86%	13%	83%	8%	73%	-2%	72%	5%	74%	10%	53%	-5%	-14%	46%	6%
Gotion	49%	0%	40%	10%	25%	5%	34%	13%	34%	0%	21%	-10%	20%	-11%	37%	25%	51%	0%	43%	10%	56%	9%	56%	0%	68%	8%	59%	4%	0%	55%	2%
EVE	31%	6%	37%	8%	37%	0%	42%	5%	40%	0%	42%	-4%	57%	-13%	65%	40%	63%	1%	67%	10%	66%	8%	73%	4%	84%	13%	78%	4%	0%	88%	11%
CALB	25%	12%	41%	22%	71%	11%	97%	6%	83%	0%	68%	-8%	95%	-10%	80%	11%	99%	13%	113%	7%	73%	-3%	104%	18%	98%	9%	67%	3%	-3%	79%	19%
Rept	122%	2%	131%	4%	138%	3%	143%	12%	65%	-7%	61%	-19%	84%	0%	54%	12%	78%	11%	69%	5%	49%	6%	31%	2%	39%	9%	31%	-2%	-4%	30%	2%
Total (top 6)	40%	8%	43%	7%	42%	10%	50%	9%	45%	-2%	39%	-6%	34%	-12%	46%	18%	58%	11%	63%	8%	59%	0%	64%	6%	67%	10%	57%	0%	-7%	53%	7%
NCM (top 2)	29%	5%	35%	4%	39%	3%	52%	3%	49%	-2%	40%	-6%	19%	-20%	33%	18%	37%	3%	31%	13%	10%	-12%	14%	9%	13%	4%	-26%	-32%	-26%	-22%	9%
LFP (top 2)	42%	10%	42%	4%	37%	14%	42%	10%	40%	-3%	36%	-5%	29%	-11%	43%	17%	58%	15%	69%	6%	74%	1%	79%	5%	82%	11%	82%	5%	-6%	68%	5%

Source: ZE Consulting, J.P. Morgan.

Figure 6: Historical Oct vs. Sep battery production volume growth (m/m %)



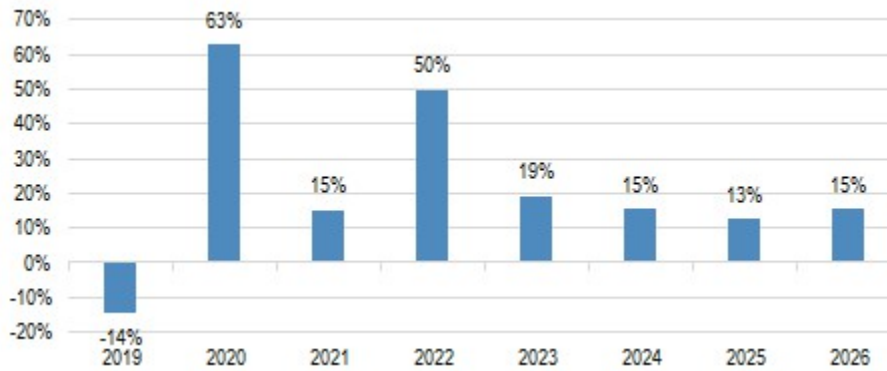
Source: ZE Consulting, J.P. Morgan. Note: Based on battery production data of top 6 battery makers: BYD, CATL, Gotion, EVE, CALB, REPT.

Table 2: Summary of production trends (quarterly/semi-annually)

	1Q25		2Q25		3Q25		4Q25		1Q26		2Q26		3Q26			1H24		2H24		FY24		1H25		2H25		FY25		1H26		YTD26	
	Y/Y	Q/Q	Y/Y	Q/Q	Y/Y	Q/Q	Y/Y	Q/Q	Y/Y	Q/Q	Y/Y	Q/Q	Y/Y	Q/Q		Y/Y	h/h	Y/Y	h/h	Y/Y	Y/Y	h/h	Y/Y	h/h	Y/Y	h/h	Y/Y	Y/Y	h/h	Y/Y	
Battery maker																															
BYD	109%	-3%	55%	6%	33%	-6%	8%	13%	-6%	-16%	15%	29%	49%	21%		36%	7%	64%	53%	52%	77%	16%	19%	2%	42%	5%	5%	3%	23%		
CATL	63%	-4%	40%	8%	43%	20%	59%	28%	58%	-5%	80%	23%	66%	11%		20%	-10%	27%	41%	24%	50%	7%	52%	42%	51%	69%	19%	65%			
Gotion	121%	-3%	81%	2%	59%	8%	31%	21%	26%	-6%	50%	22%	61%	16%		54%	7%	82%	70%	70%	99%	17%	42%	21%	63%	38%	14%	47%			
EVE	51%	-9%	29%	20%	33%	15%	40%	11%	55%	1%	65%	28%	78%	24%		92%	-8%	24%	34%	46%	38%	3%	37%	33%	37%	61%	21%	71%			
CALB	61%	-8%	31%	13%	35%	30%	83%	35%	80%	-10%	94%	22%	87%	26%		48%	0%	48%	47%	48%	44%	-2%	59%	63%	53%	87%	15%	86%			
Rept	149%	6%	64%	27%	128%	39%	109%	12%	65%	-16%	64%	26%	33%	13%		48%	0%				93%	32%	118%	65%	108%	64%	0%	48%			
Total (top 6)	79%	-4%	45%	9%	43%	14%	46%	23%	40%	-8%	60%	24%	62%	15%		31%	-4%	39%	45%	36%	60%	10%	45%	32%	51%	50%	14%	55%			
NCM (top 2)	31%	-4%	44%	17%	37%	20%	47%	9%	31%	-14%	25%	12%	0%	-4%		11%	-9%	20%	31%	16%	38%	5%	42%	35%	40%	28%	-5%	12%			
LFP (top 2)	96%	-3%	45%	4%	41%	8%	40%	28%	36%	-6%	67%	28%	81%	17%		29%	-4%	44%	49%	38%	66%	11%	40%	26%	51%	52%	21%	63%			

Source: ZE Consulting, J.P. Morgan.

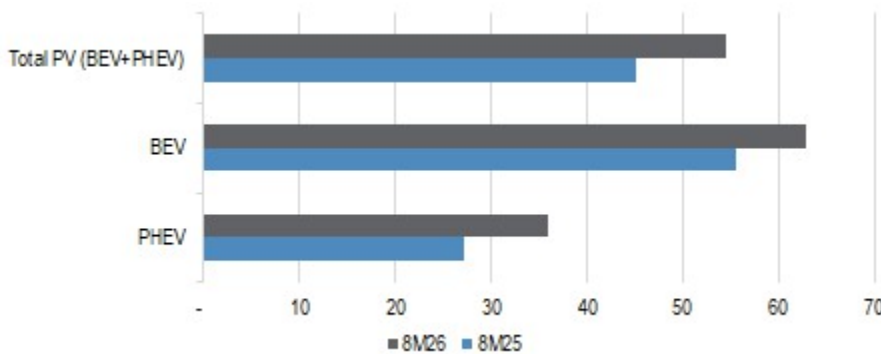
Figure 7: Historical 3Q average vs. 2Q average production volume growth (q/q %)



Source: ZE Consulting, J.P. Morgan. Note: Based on battery production data of top 6 battery makers: BYD, CATL, Gotion, EVE, CALB, REPT.

Figure 8: Average battery content per passenger vehicle in China: +21% y/y in 8M26

kWh/car



Source: ZE Consulting, GGII, J.P. Morgan.

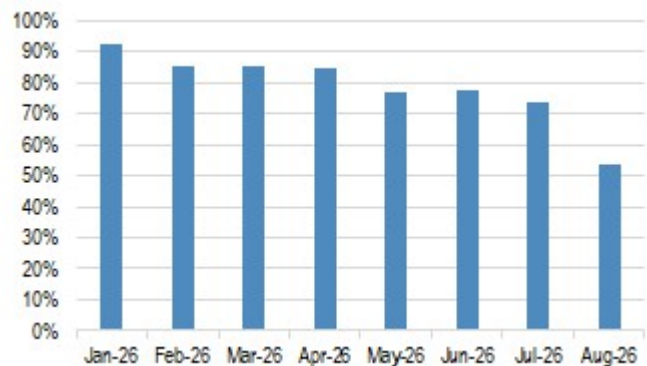
CATL's battery shipments to Li Auto

Figure 9: CATL's battery shipments to Li Auto



Source: GGII, J.P.Morgan

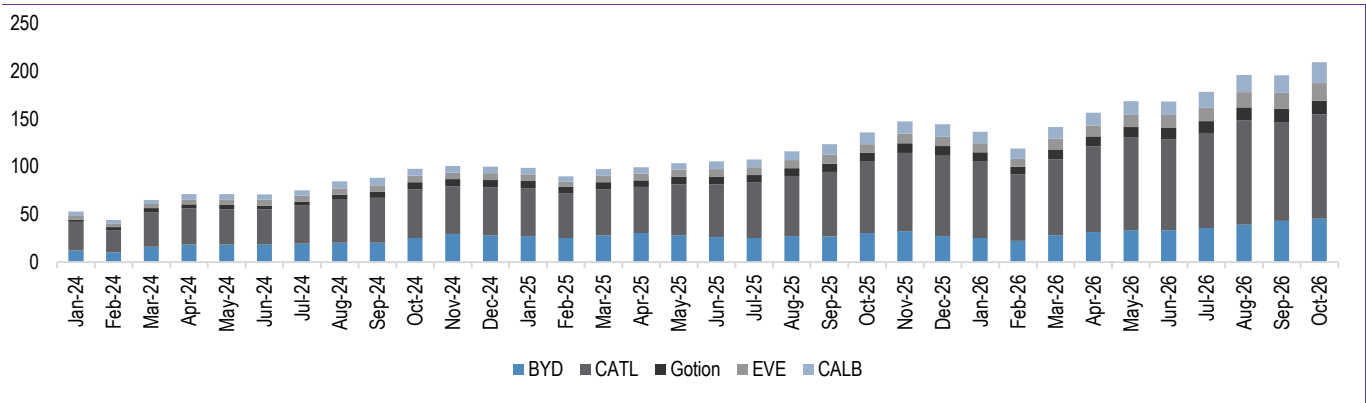
Figure 10: CATL's share within Li Auto



Source: GGII, J.P.Morgan

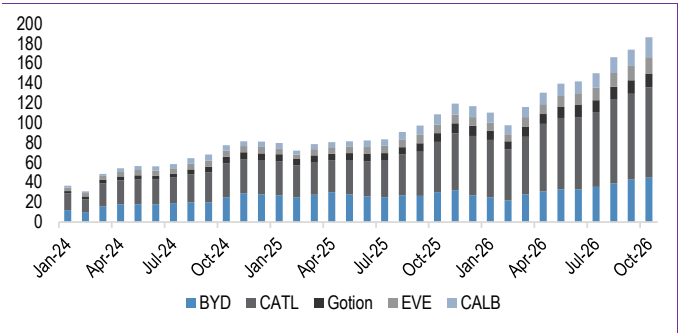
China battery & material production trend

Figure 11: Selected battery makers' total production trends (GWh)



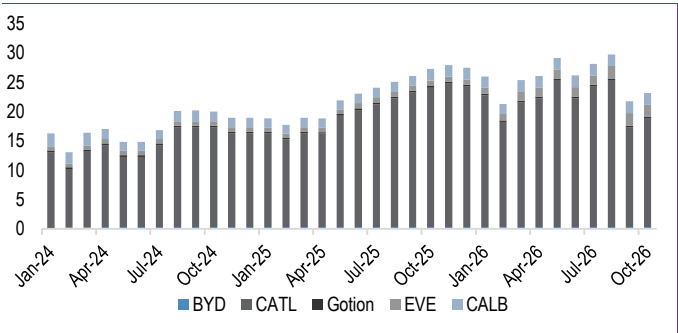
Source: ZE Consulting, GGII, J.P. Morgan. Data of the last month are based on ZE Consulting estimates.

Figure 12: Selected battery makers' LFP production trends (GWh)



Source: ZE Consulting, GGII, J.P. Morgan. Data of the last month are based on ZE Consulting estimates.

Figure 13: Selected battery makers' NCM production trends (GWh)



Source: ZE Consulting, GGII, J.P. Morgan. Data of the last month are based on ZE Consulting estimates.

Global NEV Sales Trend

Table 3: EV+PHEV monthly penetration trend by region (monthly)

K units (cars sold), % (penetration)

	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26
EV+PHEV mix															
EU-5	26%	25%	27%	29%	29%	31%	33%	28%	28%	30%	31%	32%	34%	36%	39%
China	53%	54%	54%	58%	57%	59%	59%	39%	45%	52%	61%	63%	63%	65%	65%
U.S.	10%	11%	12%	14%	7%	6%	7%	7%	7%	7%	7%	7%	7%	7%	7%
Total	34%	33%	35%	39%	37%	38%	37%	26%	25%	31%	33%	35%	36%	37%	38%
<i>M/M or Q/Q</i>	<i>2%</i>	<i>-1%</i>	<i>2%</i>	<i>4%</i>	<i>-2%</i>	<i>1%</i>	<i>-1%</i>	<i>-12%</i>	<i>-1%</i>	<i>5%</i>	<i>3%</i>	<i>1%</i>	<i>2%</i>	<i>0%</i>	<i>1%</i>
<i>Y/Y</i>	<i>5%</i>	<i>3%</i>	<i>2%</i>	<i>4%</i>	<i>3%</i>	<i>3%</i>	<i>4%</i>	<i>-2%</i>	<i>-4%</i>	<i>0%</i>	<i>3%</i>	<i>3%</i>	<i>2%</i>	<i>3%</i>	<i>3%</i>
EV+PHEV															
EU-5	225	187	137	263	219	236	254	184	188	346	246	254	335	283	210
China	1111	987	1100	1298	1282	1320	1337	596	464	852	850	950	1008	951	1005
U.S.	122	154	179	172	84	82	101	76	82	97	93	105	91	96	98
Total	1,457	1,328	1,416	1,733	1,585	1,638	1,692	856	734	1,295	1,189	1,309	1,434	1,330	1,313
EV+ PHEV y/y growth															
EU-5	23%	43%	44%	39%	46%	42%	34%	31%	33%	42%	41%	39%	49%	51%	53%
China	30%	12%	7%	16%	7%	4%	3%	-20%	-32%	-14%	-6%	-7%	-9%	-4%	-9%
U.S.	-7%	14%	19%	27%	-37%	-44%	-40%	-40%	-35%	-35%	-31%	-21%	-25%	-38%	-45%
Total	24%	16%	11%	20%	7%	4%	2%	-15%	-23%	-6%	-2%	-2%	-2%	0%	-7%
PV															
EU-5	868	737	507	901	757	753	777	660	677	1166	799	791	973	777	539
China	2,084	1,844	2,019	2,241	2,242	2,225	2,261	1,544	1,034	1,648	1,384	1,510	1,602	1,461	1,541
U.S.	1,280	1,406	1,480	1,267	1,284	1,292	1,488	1,114	1,196	1,405	1,380	1,480	1,379	1,380	1,388
Total	4,232	3,987	4,006	4,409	4,283	4,270	4,526	3,318	2,907	4,219	3,563	3,781	3,954	3,618	3,468
PV y/y growth															
EU-5	-6%	0%	4%	10%	5%	2%	0%	-1%	3%	11%	8%	3%	12%	5%	6%
China	17%	6%	5%	5%	-2%	-8%	-14%	-14%	-25%	-15%	-21%	-22%	-23%	-21%	-24%
U.S.	-4%	9%	4%	7%	-4%	-6%	-2%	-1%	-4%	-12%	-6%	0%	8%	-2%	-6%
Total	5%	6%	5%	7%	-1%	-6%	-8%	-8%	-12%	-8%	-10%	-9%	-7%	-9%	-13%

Source: Bloomberg Finance L.P., Industry Associations, J.P. Morgan. * EU-5 includes Germany, France, Spain, Italy and the UK. China uses CPCA retail data. Note: China Dec monthly sales are estimated by CPCA

Table 4: EV+PHEV monthly penetration trend by region (quarterly)

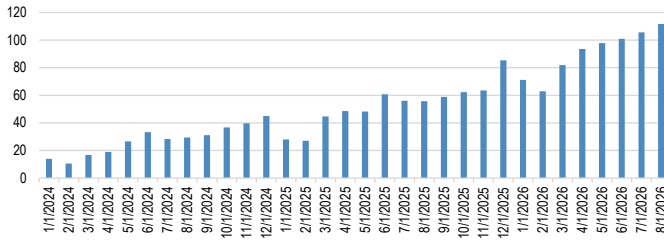
K units (cars sold), % (penetration)

	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	FY22	FY23	FY24	FY25	YTD26
EV+PHEV mix															
EU-5	18%	18%	20%	23%	22%	24%	27%	31%	29%	33%	22%	21%	20%	26%	32%
China	36%	46%	52%	51%	47%	53%	55%	59%	45%	62%	27%	35%	47%	54%	57%
U.S.	10%	10%	11%	11%	10%	9%	12%	7%	7%	7%	7%	9%	10%	10%	7%
Total	23%	27%	33%	34%	29%	32%	36%	38%	28%	35%	20%	24%	30%	34%	33%
<i>M/M or Q/Q</i>	<i>-4%</i>	<i>4%</i>	<i>6%</i>	<i>1%</i>	<i>-5%</i>	<i>3%</i>	<i>4%</i>	<i>1%</i>	<i>-10%</i>	<i>7%</i>					
<i>Y/Y</i>	<i>3%</i>	<i>6%</i>	<i>8%</i>	<i>7%</i>	<i>6%</i>	<i>5%</i>	<i>3%</i>	<i>3%</i>	<i>-2%</i>	<i>2%</i>	<i>8%</i>	<i>4%</i>	<i>6%</i>	<i>4%</i>	<i>1%</i>
EV+PHEV															
EU-5	429	449	415	505	526	581	587	709	718	835	1,728	1,870	1,798	2,403	2,046
China	1,768	2,334	3,027	3,767	2,421	3,037	3,385	3,939	1,912	2,808	5,661	7,747	10,896	12,782	6,676
U.S.	363	402	420	447	402	390	505	267	255	289	928	1,434	1,632	1,564	738
Total	2,560	3,185	3,863	4,719	3,349	4,008	4,477	4,915	2,885	3,932	8,317	11,051	14,326	16,749	9,460
EV+ PHEV y/y growth															
EU-5	7%	-1%	-15%	-3%	23%	29%	41%	41%	36%	44%	16%	8%	-4%	34%	43%
China	34%	32%	44%	47%	37%	30%	12%	5%	-21%	-8%	89%	37%	41%	17%	-12%
U.S.	16%	13%	7%	20%	11%	-3%	20%	-40%	-37%	-26%	60%	55%	14%	-4%	-34%
Total	26%	23%	29%	37%	31%	26%	16%	4%	-14%	-2%	64%	33%	30%	17%	-6%
PV															
EU-5	2,412	2,434	2,046	2,238	2,378	2,373	2,145	2,287	2,503	2,563	7,924	9,089	9,130	9,184	6,382
China	4,874	5,055	5,778	7,338	5,120	5,771	6,104	6,728	4,226	4,496	20,809	21,955	23,045	23,723	11,724
U.S.	3,787	4,118	3,901	4,220	3,960	4,221	4,153	4,064	3,715	4,239	13,853	15,633	16,027	16,398	10,722
Total	11,073	11,607	11,725	13,797	11,458	12,365	12,402	13,079	10,444	11,298	42,585	46,677	48,202	49,304	28,828
PV y/y growth															
EU-5	7%	4%	-6%	-4%	-1%	-2%	5%	2%	5%	8%	-4%	15%	0%	1%	6%
China	12%	-5%	0%	12%	5%	14%	6%	-8%	-17%	-22%	2%	6%	5%	3%	-21%
U.S.	5%	0%	-2%	8%	5%	2%	6%	-4%	-6%	0%	-8%	13%	3%	2%	-3%
Total	9%	-1%	-2%	8%	3%	7%	6%	-5%	-9%	-9%	-2%	10%	3%	2%	-9%

Source: Bloomberg Finance L.P., Industry Associations, J.P. Morgan. * EU-5 includes Germany, France, Spain, Italy and the UK. China uses CPCA retail data. Note: China Dec monthly sales are estimated by CPCA

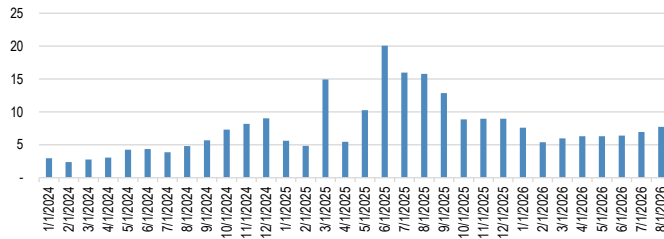
Chinese players' ESS battery shipments in different regions

Figure 14: Global ESS battery shipments (monthly) (Unit: Gwh)



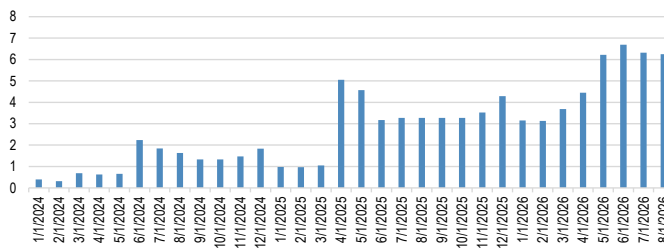
Source: ICCSino

Figure 16: Chinese players' ESS battery exports to the US (monthly) (Unit: Gwh)



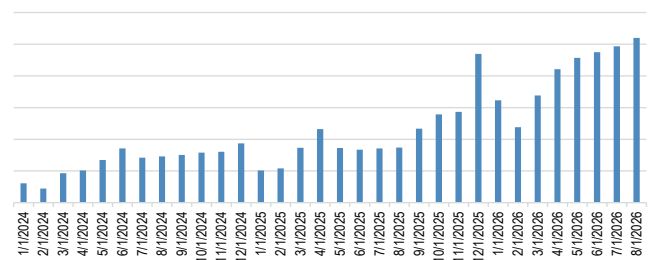
Source: ICCSino

Figure 18: Chinese players' ESS battery exports to the EU residential market (monthly) (Unit: Gwh)



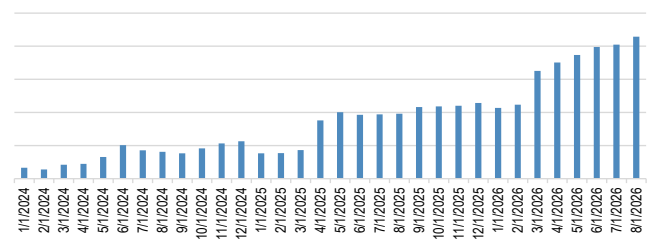
Source: ICCSino.

Figure 15: China domestic demand ESS shipments (monthly) (Unit: Gwh)



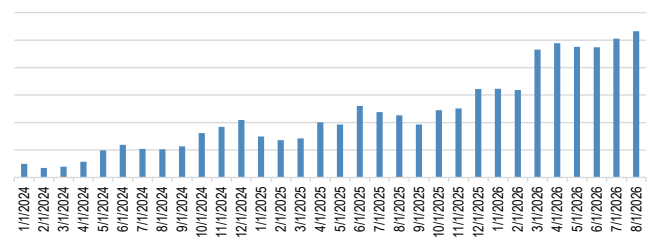
Source: ICCSino

Figure 17: Chinese players' ESS battery exports to the EU (monthly) (Unit: Gwh)



Source: ICCSino

Figure 19: Chinese players' ESS battery exports to RoW (markets excluding China, the US and the EU) (monthly) (Unit: Gwh)



Source: ICCSino.

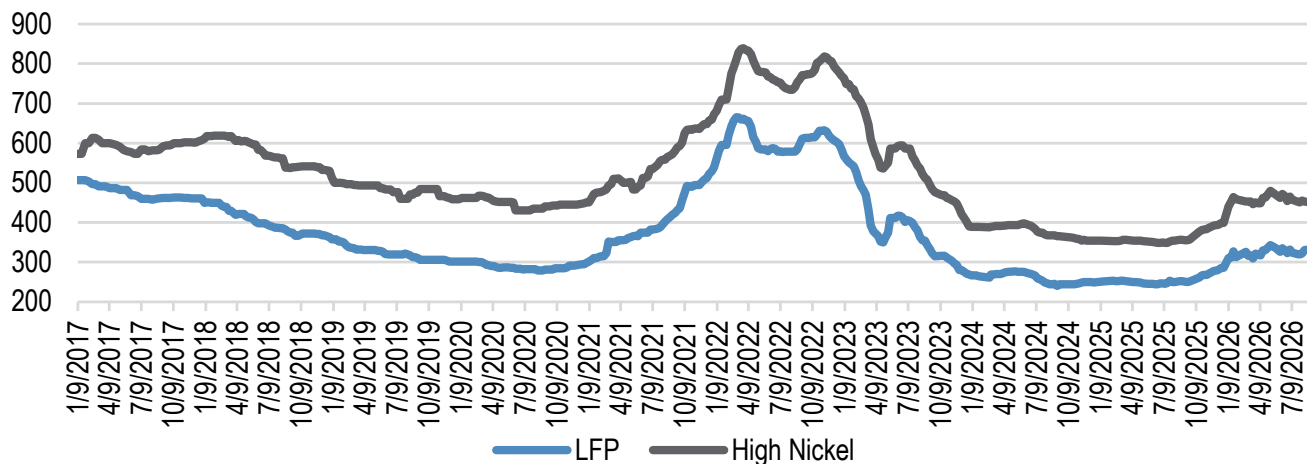
Summary of battery and material prices

Table 5: Summary of battery and material price changes (market spot price changes)

	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26 so far
Battery cell											
Consumer (mid-end cylindrical)	-	-	-	-	-	-	-	-	-	-	-
Consumer (high-end cylindrical)	-	-	-	-	-	-	-	-	-	-	-
EV (NCM prismatic)	-10%	-3%	-1%	-	-2%	-5%	-	4%	9%	-	-
EV (LFP prismatic)	-13%	-5%	-2%	-	-4%	-9%	-	-	9%	-	-
EV (18650 cylindrical)	-	-	-	-	-	-	-	-	-	-	-
ESS (CH player)	-	-	-	-	-1%	-4%	7%	-	10%	6%	3%
Battery material											
LFP cathode	0.20%	-10%	-17%	-2%	-	-10%	13%	31%	16%	-2%	-10%
NCM cathode (NCM523)	2%	-11%	-4%	-2%	18%	-6%	3%	30%	18%	1%	-11%
NCM cathode (NCM811)	1%	-6%	-1%	-1%	5%	-3%	2%	16%	17%	3%	-8%
Artificial graphite anode	-6%	-	-	-	1%	1%	-1%	-	-3%	-	-
Natural graphite anode	-5%	-2%	-	-	-	-	-	-	-	-	-
Separator (Wet 9µm: mid-end)	-20%	-2%	-	-	-7%	-8%	3%	17%	-	4%	-
Separator (Dry 16µm: mid-end)	-23%	-	-5%	-	5%	-	-	-	-	-	-
NCM Electrolyte	-6%	-18%	-1%	2%	-9%	-5%	2%	66%	-16%	-6%	12%
LFP Electrolyte	-3%	-19%	-3%	3%	-10%	-6%	2%	93%	-17%	-6%	18%
Copper foil processing fee (5µm)	-	-	-	-	-4%	-12%	-9%	14%	-	4%	4%
Copper foil processing fee (6µm)	-	9%	-11%	15%	-	-	-	-	5%	5%	10%
Raw material											
Lithium carbonate (mainly for LFP)	21%	-15%	-17%	-	-3%	-16%	19%	63%	32%	-8%	-10%
Lithium hydroxide (mainly for NCM)	12%	-18%	-14%	-1%	-3%	-15%	26%	40%	52%	-11%	-14%
Cobalt	1%	-5%	-13%	5%	37%	-3%	5%	50%	4%	-2%	-31%
Nickel	3%	2%	-2%	-6%	3%	-5%	-0.40%	8%	1%	-6%	0%
LiPF6	-1%	-11%	-13%	13%	-3%	-15%	18%	195%	-41%	-1%	11%

Source: WIND, ICCSino, SMM, J.P. Morgan. Note: "-" means no changes. Note: LFP cathode price changes since 2Q26 refer to LFP premium (EV) cathode price changes. 3Q26 so far refers to July 1- Sep 25.

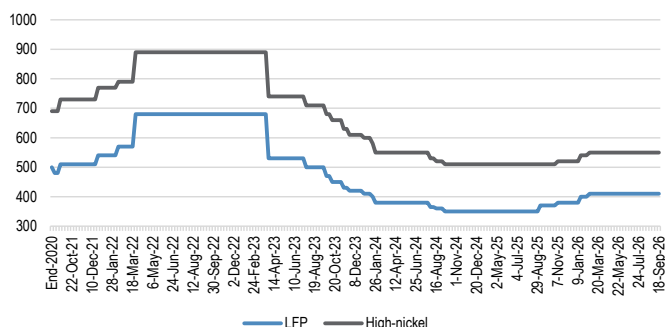
Figure 20: Estimated battery material cost (Rmb/kWh)



Source: ZE Consulting, J.P. Morgan.

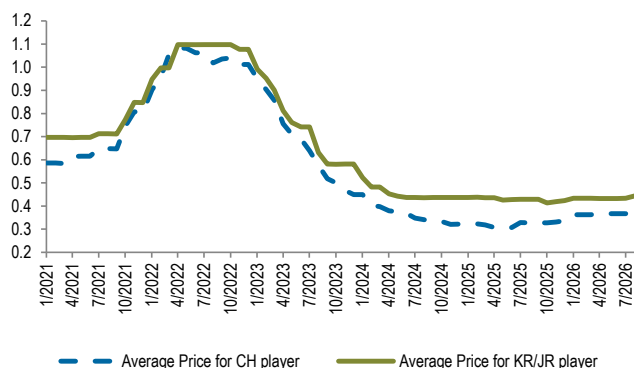
Battery prices

Figure 21: China EV battery price at cell level (excl. VAT, Rmb/kWh), as per ZE Consulting



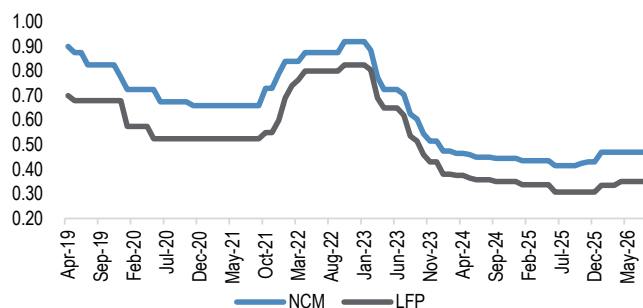
Source: ZE Consulting, J.P. Morgan.

Figure 23: ESS battery prices: CH vs. KR/JP players



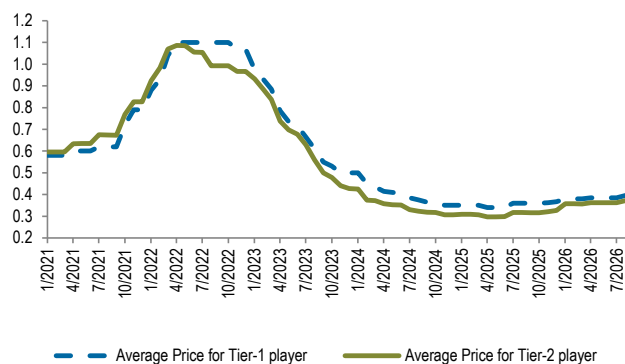
Source: ICCSino, J.P. Morgan. Note: Prices include VAT and are based on new contract prices.

Figure 22: China EV battery price at cell level (excl. VAT, Rmb/Wh), as per CIAPS



Source: CIAPS, ICCSino, J.P. Morgan.

Figure 24: ESS battery prices: Chinese tier-1 vs. tier-2

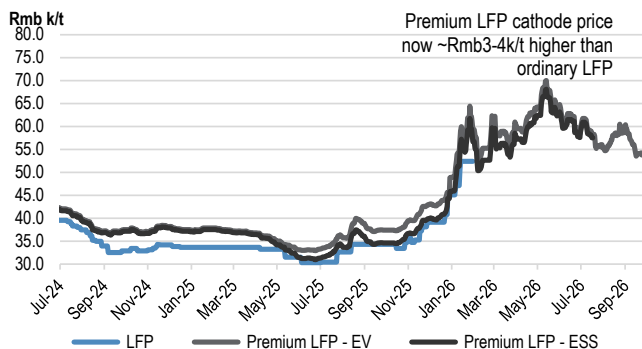


Source: ICCSino, J.P. Morgan. Note: Prices include VAT and are based on new contract prices.

Battery material prices

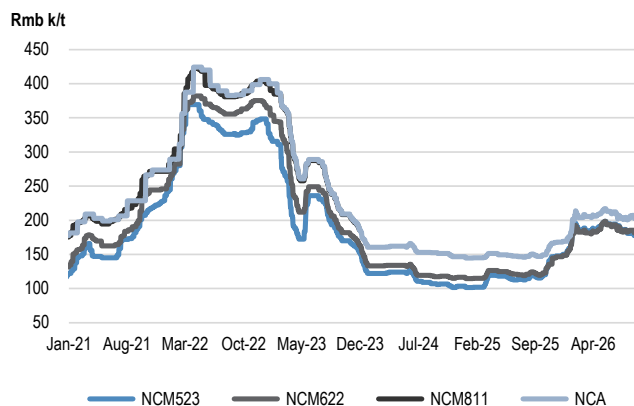
Battery material

Figure 25: LFP cathode price (Rmb10k/ton)



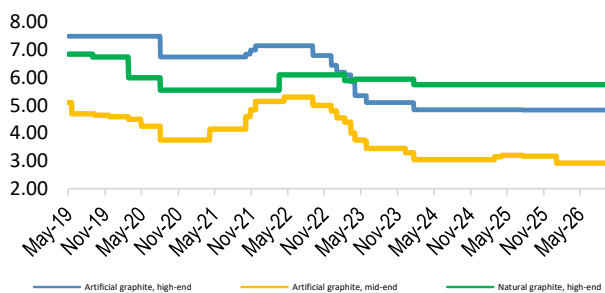
Source: Wind, ICCSino, J.P. Morgan.

Figure 26: NCM cathode prices (Rmb10k/ton)



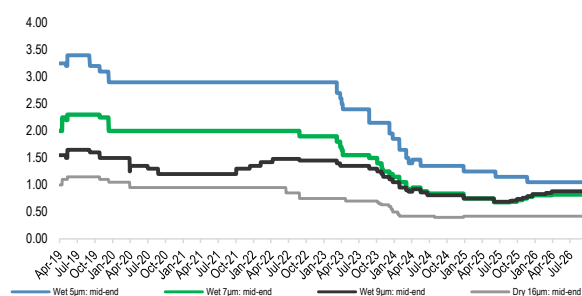
Source: Wind, ICCSino, J.P. Morgan.

Figure 27: Anode prices (Rmb10k/ton)



Source: Wind, ICCSino, J.P. Morgan.

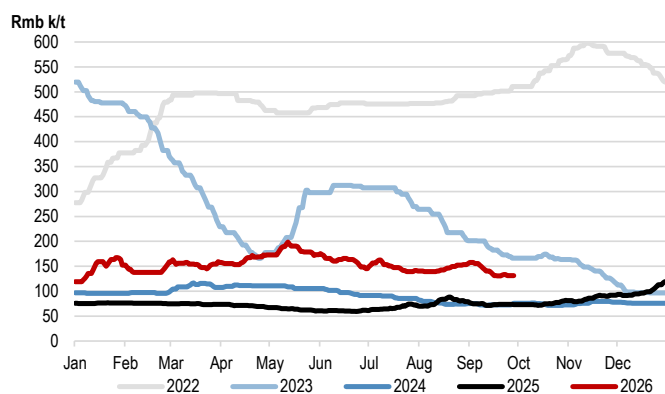
Figure 29: Separator prices (Rmb/sqm)



Source: Wind, ICCSino, J.P. Morgan.

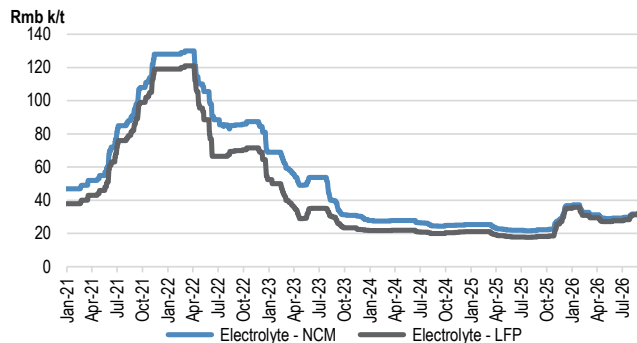
Raw materials

Figure 31: Lithium carbonate price



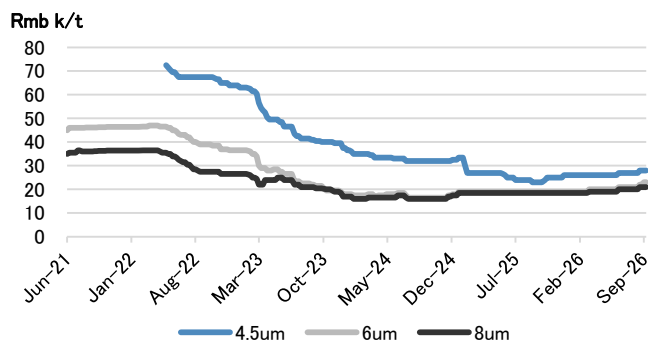
Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 28: Electrolyte prices (Rmb10k/ton)



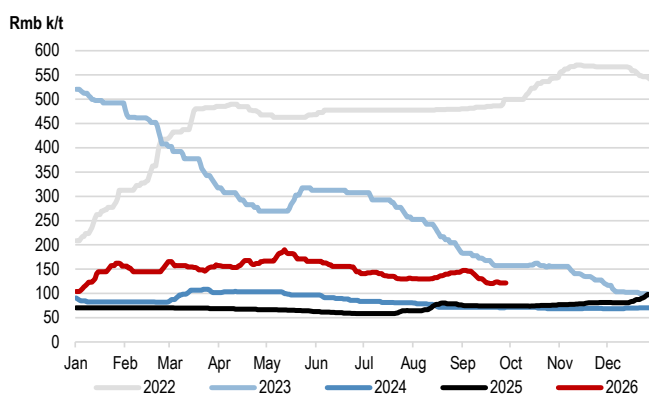
Source: SMM, ICCSino, J.P. Morgan.

Figure 30: Copper foil processing fee (Rmb k/ton)



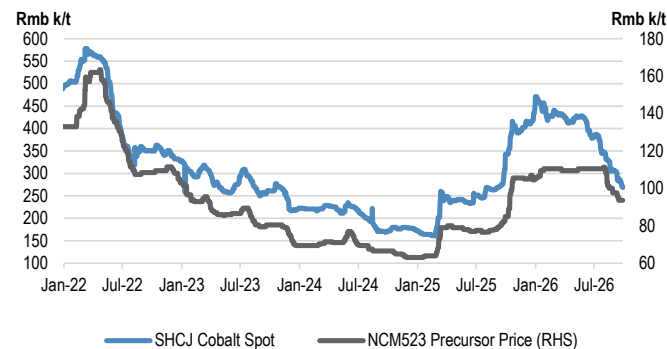
Source: SMM, J.P. Morgan.

Figure 32: Lithium hydroxide price



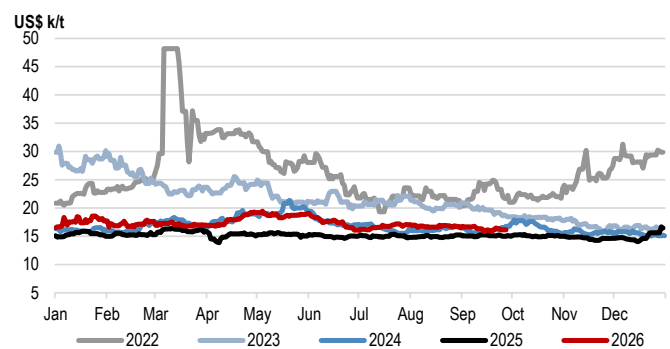
Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 33: Cobalt and precursor prices



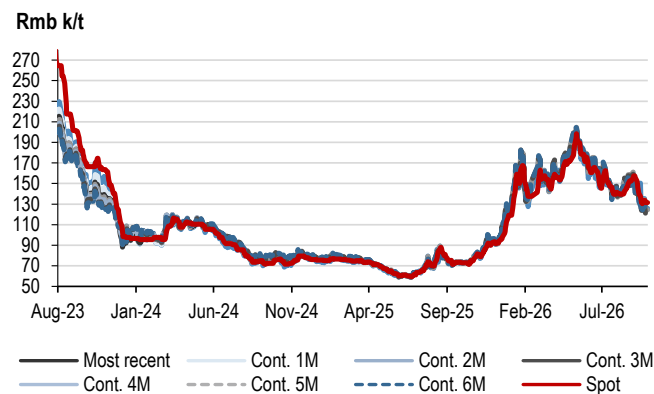
Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 34: LME nickel price



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 35: Lithium GZ future price (Rmb k/t)



Source: Bloomberg Finance L.P., J.P. Morgan.

Valuation comparison

Table 6: Valuation comparison of China battery supply chain

			Cover	Price	Mkt cap	P/E (x)		EV/EBITDA (x)		Dividend yield		ROE		Price change
Company	Code	Rec	analyst	28-Sep	(US\$M)	26E	27E	26E	27E	26E	27E	26E	27E	YTD
Upstream - raw material														
Tianqi Lithium	002466 CH	N	Avery Chan	39.4	9,528	9.6	8.5	6.2	5.5	2%	2%	19%	18%	-34%
Ganfeng Lithium	002460 CH	OW	Avery Chan	41.9	11,907	9.7	8.8	3.6	3.2	2%	2%	15%	14%	-29%
CMOC	3993 HK	OW	Avery Chan	14.9	51,233	8.8	7.9	5.5	4.8	3%	4%	33%	29%	-23%
Zhejiang Huayou	603799 CH	NC	-	34.1	9,629	8.4	6.5	na	na	2%	2%	15%	16%	-50%
Midstream - battery material														
Beijing Easpring	300073 CH	NC	-	36.3	2,944	17.7	13.6	8.6	7.1	1%	2%	7%	9%	-37%
Shanghai Putailai	603659 CH	OW	Rebecca Wen	20.8	6,643	13.7	9.6	9.1	7.2	1%	1%	14%	16%	-24%
Shanshan	600884 CH	NC	-	11.2	3,743	nm	nm	na	na	1%	1%	6%	7%	-17%
Shangtai	001301 CH	NC	-	46.8	2,541	15.3	8.3	na	na	2%	2%	14%	22%	-24%
Yunnan Energy	002812 CH	N	Rebecca Wen	45.1	6,591	24.7	13.5	na	na	0%	0%	8%	13%	-20%
Capchem	300037 CH	NC	-	66.7	7,487	22.4	18.4	16.1	13.0	1%	2%	18%	19%	27%
Do-Fluoride	002407 CH	NC	-	29.8	5,275	16.7	17.0	na	na	3%	3%	12%	11%	-12%
Shenzhen Senior	300568 CH	NC	-	16.1	3,365	33.2	18.9	na	na	1%	1%	7%	10%	6%
Hunan Yuneng	301358 CH	N	Rebecca Wen	49.5	6,250	9.6	8.5	7.1	6.8	1%	1%	26%	24%	-23%
Dynanonic	300769 CH	UW	Rebecca Wen	38.1	1,591	16.6	13.9	na	na	0%	0%	14%	15%	-13%
Ningbo Ronbay	688005 CH	UW	Rebecca Wen	22.4	2,389	40.7	27.0	15.5	11.5	0%	0%	4%	8%	-37%
CNGR	300919 CH	OW	Parsley Ong	36.2	5,339	10.7	n/a	8.1	7.4	3%	0%	11%	12%	-22%
Battery equipment														
Wuxi Lead-A	300450 CH	OW	Rebecca Wen	31.2	7,628	18.3	13.3	16.8	12.4	2%	2%	16%	19%	-38%
Wuxi Lead-H	470 HK	OW	Rebecca Wen	25.5	7,628	12.8	9.3	16.8	12.4	2%	3%	16%	19%	-44%
Yinghe	300457 CH	NC	-	20.3	1,948	19.7	16.3	12.6	10.7	1%	1%	9%	10%	-27%
Hangke	688006 CH	NC	-	19.9	1,788	22.6	15.8	na	na	1%	1%	8%	11%	-32%
Downstream cell makers														
Gotion	002074 CH	UW	Rebecca Wen	26.1	7,047	23.2	17.0	9.2	na	1%	1%	7%	9%	-33%
CATL-A	300750 CH	OW	Rebecca Wen	292.0	205,244	14.2	11.5	7.8	6.1	3%	4%	26%	26%	-20%
CATL-H	3750 HK	OW	Rebecca Wen	484.6	205,244	20.1	16.3	7.8	6.1	2%	3%	26%	26%	-3%
EVE Energy	300014 CH	N	Rebecca Wen	47.6	15,403	14.3	11.6	10.4	8.3	2%	2%	15%	16%	-28%
CALB	3931 HK	OW	Rebecca Wen	17.2	3,895	9.8	6.9	6.8	5.5	0%	0%	7%	9%	-31%
REPT	666 HK	NC	-	8.7	2,589	13.1	9.9	6.0	4.9	na	na	11%	13%	-35%
Sunwoda	300207 CH	NC	-	21.4	5,883	19.7	13.2	7.3	5.7	1%	2%	8%	11%	-18%
NEV OEMs														
Brilliance China	1114 HK	N	Nick Lai	1.8	1,167	4.4	3.9	na	na	32%	11%	9%	10%	-55%
Great Wall Motor	2333 HK	UW	Nick Lai	7.4	15,019	6.6	6.0	5.4	4.6	6%	5%	9%	9%	-52%
Geely Automobile	175 HK	OW	Nick Lai	16.2	22,233	6.5	5.2	2.4	2.0	3%	4%	21%	23%	-10%
GAC	2238 HK	OW	Nick Lai	2.5	6,507	nm	nm	na	na	0%	0%	-5%	-1%	-38%
SAIC	600104 CH	UW	Nick Lai	10.8	18,441	12.3	10.9	1.4	1.1	2%	2%	3%	4%	-29%
BAIC Motor-H	1958 HK	N	Nick Lai	0.7	761	nm	nm	na	na	0%	0%	-5%	-1%	-61%
Changan Auto	200625 CH	UW	Nick Lai	3.2	9,362	17.0	13.5	4.5	3.4	4%	2%	2%	3%	-24%
BYD	1211 HK	OW	Nick Lai	77.7	103,945	14.1	11.0	na	na	1%	1%	16%	18%	-19%
NIO Inc - Adr	NIO US	N	Nick Lai	3.6	9,121	nm	nm	7.6	4.6	0%	0%	-55%	-19%	-29%
XPeng Inc - Adr	XPEV US	OW	Nick Lai	10.1	9,637	nm	nm	nm	nm	nm	nm	-16%	-4%	-50%
Leapmotor	9863 HK	OW	Nick Lai	35.6	6,141	16.3	9.4	4.9	0.6	0%	0%	17%	23%	-27%
Li Auto - Adr	LI US	UW	Nick Lai	11.8	12,172	nm	50.2	na	2.0	0%	0%	-7%	2%	-31%
Auto parts														
Mint	425 HK	OW	Rebecca Wen	23.5	3,509	7.8	6.6	na	4.2	4%	5%	12%	13%	-26%
Nexteer	1316 HK	N	Rebecca Wen	4.6	1,470	8.5	7.0	2.0	1.9	3%	5%	8%	9%	-28%
Fuyao-H	3606 HK	N	Rebecca Wen	55.0	20,040	13.6	11.4	9.6	8.2	4%	5%	23%	25%	-18%
Fuyao-A	600660 CH	N	Rebecca Wen	52.9	20,040	15.3	12.8	9.6	8.2	4%	5%	23%	25%	-18%
Huayu	600741 CH	NC	-	15.0	7,035	6.5	6.0	1.9	3.5	6%	6%	10%	11%	-25%
Joyson	600699 CH	N	Rebecca Wen	24.0	5,233	21.3	17.5	4.0	3.3	1%	1%	9%	10%	-23%
Tuopu	601689 CH	UW	Rebecca Wen	46.5	12,037	29.0	25.5	na	na	1%	1%	11%	12%	-40%
Sanhua	002050 CH	OW	Karen Li	35.4	21,154	34.9	27.6	19.6	16.7	1%	1%	13%	14%	-36%
Sanhua-H	2050 HK	OW	Karen Li	24.0	21,154	23.6	18.7	19.6	16.7	1%	2%	n/a	n/a	-37%
Simple average														
						15.3	13.0	8.9	7.1	2%	2%	12%	14%	-28%

Source: Bloomberg Finance L.P., J.P. Morgan estimates. Note: Bloomberg consensus estimates for Not Covered (NC) stocks, J.P. Morgan estimates for all others. "nm" = not meaningful (loss or near breakeven). EV/EBITDA (x) is from the Bloomberg consensus.

Companies Discussed in This Report (all prices in this report as of market close on 28 September 2026, unless otherwise indicated)

CATL - A(300750.SZ/Rmb291.99/OW), CATL - H(3750.HK/HK\$484.60/OW)

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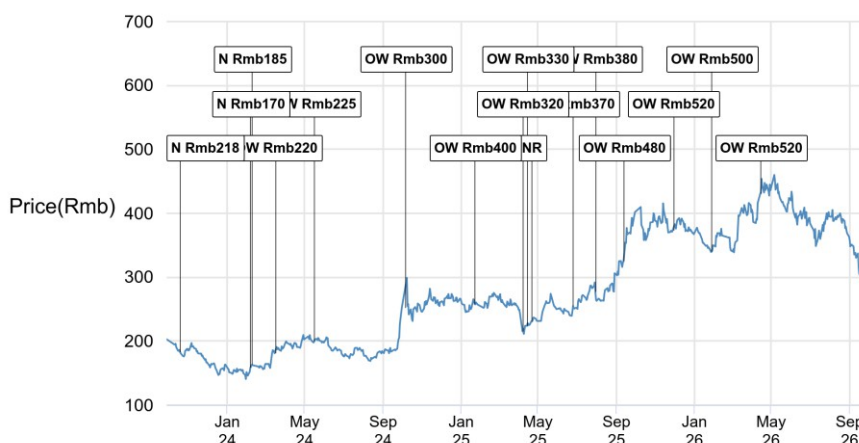
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CATL - A (300750.SZ, 300750 CH) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Sep 27, 2018. All share prices are as of market close on the previous business day.

CATL - H (3750.HK, 3750 HK) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 25, 2025. All share prices are as of market close on the previous business day.

Date	Rating	Price (Rmb)	Price Target (Rmb)
20-Oct-23	N	182.70	218
07-Feb-24	N	158.33	170
09-Feb-24	N	162.84	185
17-Mar-24	OW	181.00	220
17-May-24	OW	199.67	225
07-Oct-24	OW	251.89	300
23-Jan-25	OW	258.00	400
08-Apr-25	OW	215.22	320
15-Apr-25	OW	224.00	330
22-Apr-25	NR	231.36	--
26-Jun-25	OW	254.90	370
31-Jul-25	OW	277.09	380
14-Sep-25	OW	325.00	480
30-Nov-25	OW	373.20	520
29-Jan-26	OW	339.99	500
16-Apr-26	OW	431.00	520

Date	Rating	Price (HK\$)	Price Target (HK\$)
26-Jun-25	OW	319.00	400
31-Jul-25	N	426.60	415
14-Sep-25	OW	432.80	530
05-Oct-25	N	610.50	600
16-Nov-25	N	542.00	575
30-Nov-25	OW	472.00	650
29-Jan-26	OW	475.40	640
19-Mar-26	N	650.50	650
16-Apr-26	OW	655.50	725

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